Building your dreams with GenAI

Now is the best time ever to build your ideas

Let me start with some examples

- sitinshade.com
- some kerlaa stuff

Agenda

- looking at problem statements
- breaking down real life projects

Why should you care?

- you'll be thinking about problems in a differnet way
- you'll be so much more equipped with gen-ai
- you'll be in a much better state to use gen-ai to go start building your ideasQ

Aldrin Jenson

- Product Engineer at **Athena Intelligence**
- R&D at **Thoughtworks Technologies**
- Alumni from **Govt. Model Engineering College**

@aldrinjenson

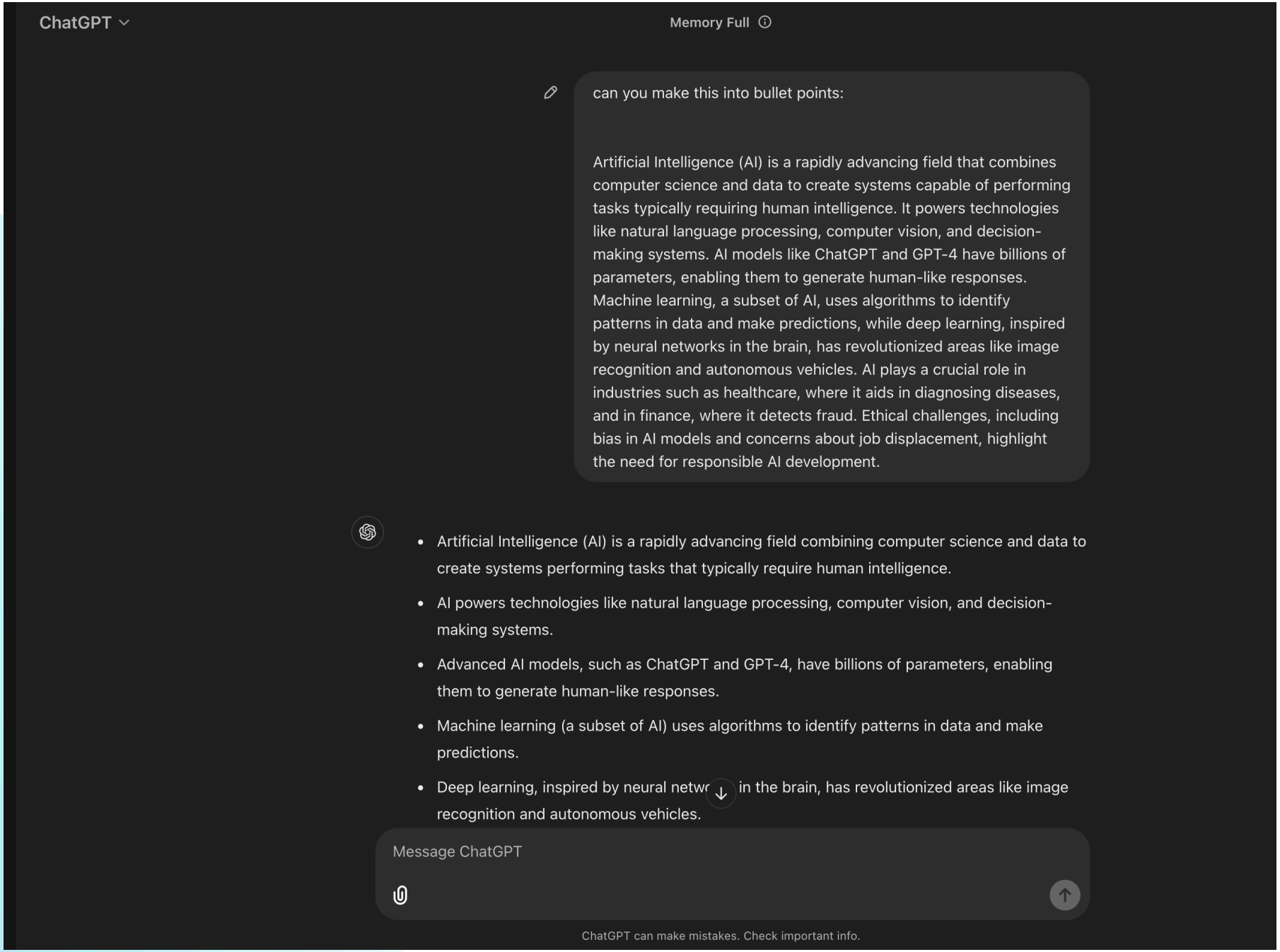
let's begin.

Problem Statement 1

Making Slides is hard

What is the actual process?

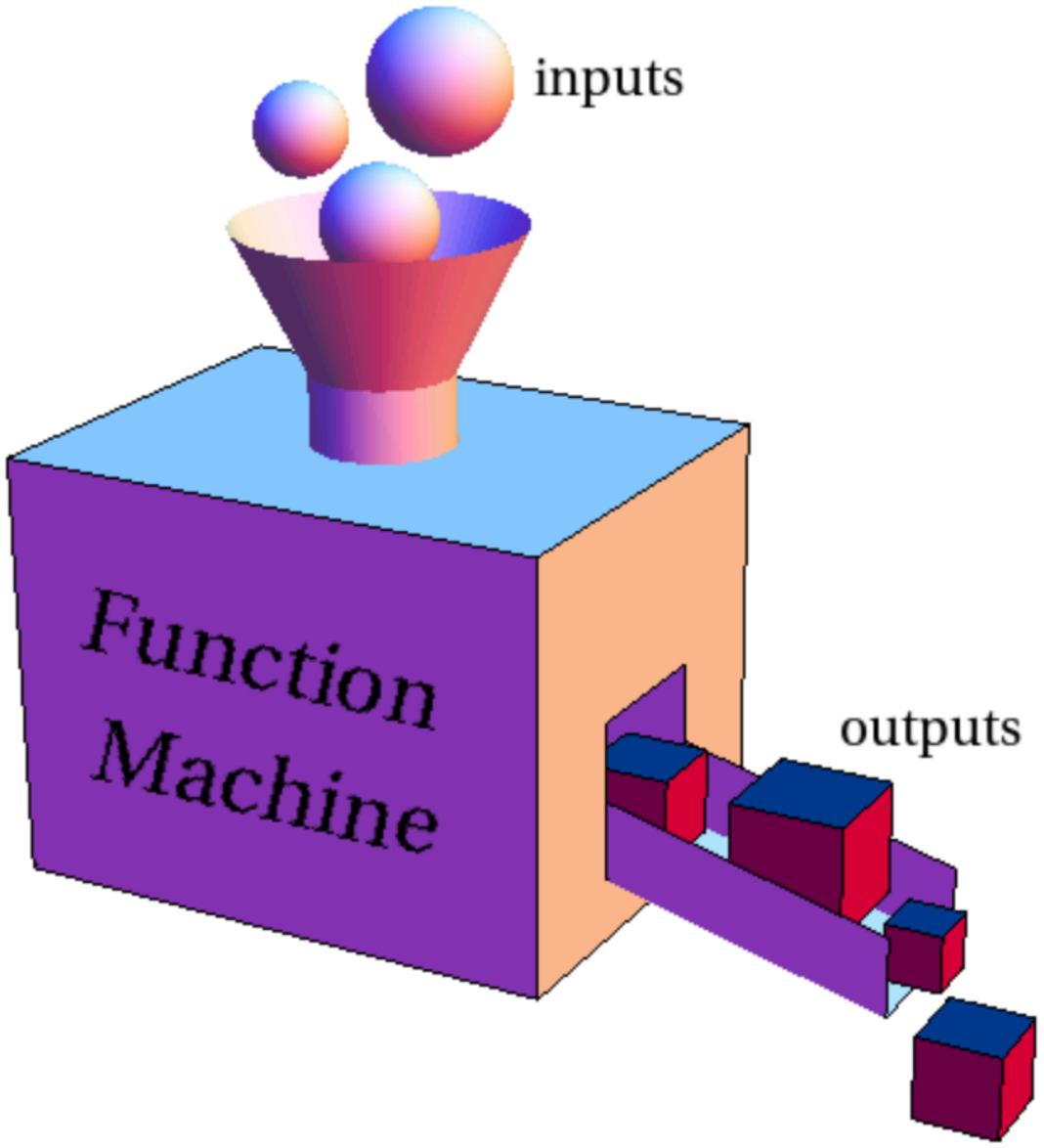
How would you solve this??

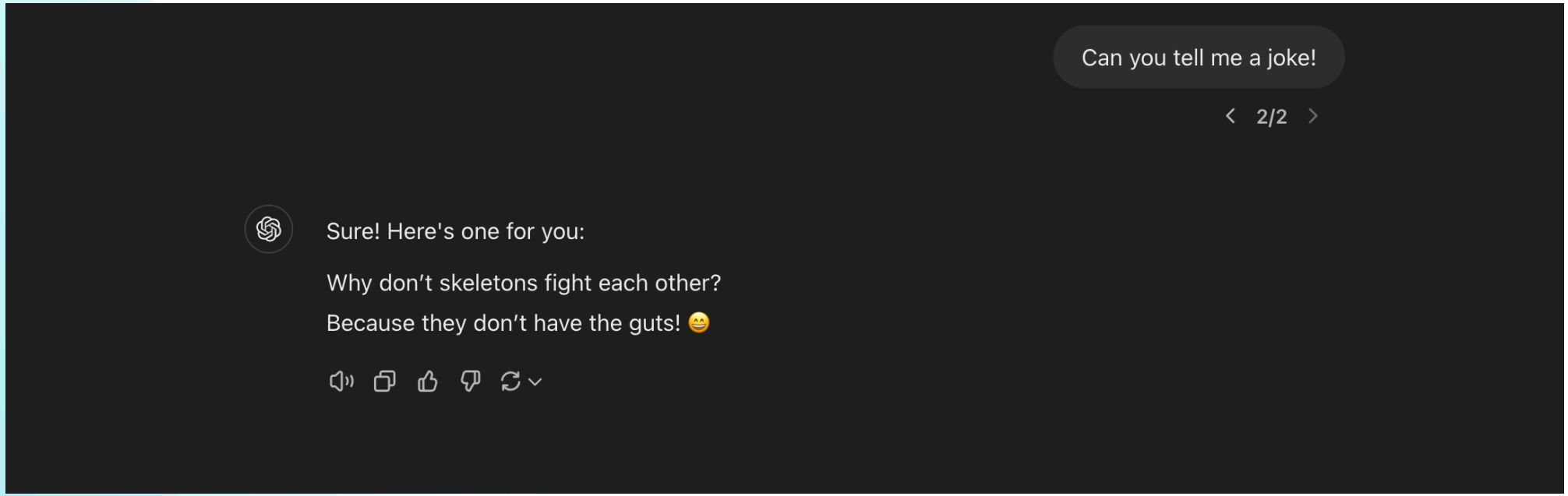

ChatGPT
Memory Full
can you make this into bullet points:
Artificial Intelligence (AI) is a rapidly advancing field that combines computer science and data to create systems capable of performing tasks typically requiring human intelligence. It powers technologies like natural language processing, computer vision, and decision-making systems. AI models like ChatGPT and GPT-4 have billions of parameters, enabling them to generate human-like responses. Machine learning, a subset of AI, uses algorithms to identify patterns in data and make predictions, while deep learning, inspired by neural networks in the brain, has revolutionized areas like image recognition and autonomous vehicles. AI plays a crucial role in industries such as healthcare, where it aids in diagnosing diseases, and in finance, where it detects fraud. Ethical challenges, including bias in AI models and concerns about job displacement, highlight the need for responsible AI development.
• Artificial Intelligence (AI) is a rapidly advancing field combining computer science and data to create systems performing tasks that typically require human intelligence.
• AI powers technologies like natural language processing, computer vision, and decision-making systems.
• Advanced AI models, such as ChatGPT and GPT-4, have billions of parameters, enabling them to generate human-like responses.
• Machine learning (a subset of AI) uses algorithms to identify patterns in data and make predictions.
• Deep learning, inspired by neural netw in the brain, has revolutionized areas like image recognition and autonomous vehicles.
Message ChatGPT
ChatGPT can make mistakes. Check important info.

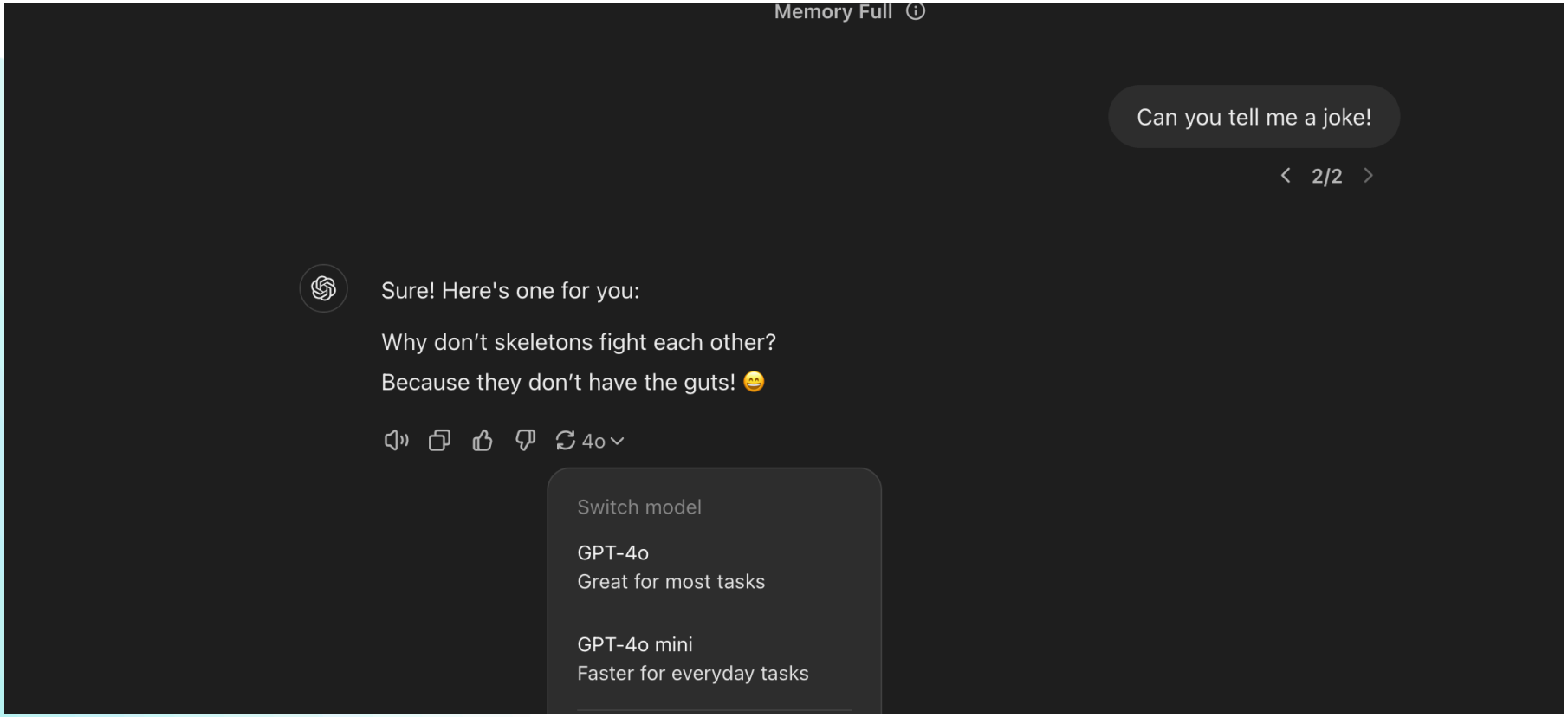
What exactly is chatGPT?

- an AI?
- a Large language Model?
- some Framework?

A Function


inputs
Function
Machine
outputs


Can you tell me a joke!
2/2
Sure! Here's one for you:
Why don't skeletons fight each other?
Because they don't have the guts!


Memory Full
Can you tell me a joke!
2/2
Sure! Here's one for you:
Why don't skeletons fight each other?
Because they don't have the guts! 😄
4o
Switch model
GPT-4o
Great for most tasks
GPT-4o mini
Faster for everyday tasks

prepare a travel plan for visiting Trivandrum

Here's a 2-day travel plan for exploring Trivandrum (Thiruvananthapuram), Kerala, known for its cultural heritage, beautiful beaches, and spiritual sites.

Day 1: Exploring the Cultural and Spiritual Side

Morning

- **Sree Padmanabhaswamy Temple**
  - Start your day with a visit to this iconic temple dedicated to Lord Vishnu. Admire its architectural beauty and spiritual ambiance.
  - Dress code: Traditional attire (mandatory).
- **Kuthiramalika Palace Museum** (near the temple)
  - Explore this 19th-century palace showcasing Travancore royal history, exquisite wooden carvings, and antique artifacts.

Afternoon

- **Lunch**: Enjoy traditional Kerala cuisine at a nearby restaurant like Mother's Veg Plaza.
- **Napier Museum and Zoo**
  - Visit the museum for its eclectic collection of artifacts and the lush zoological park featuring a variety of animals and birds.

Markdown

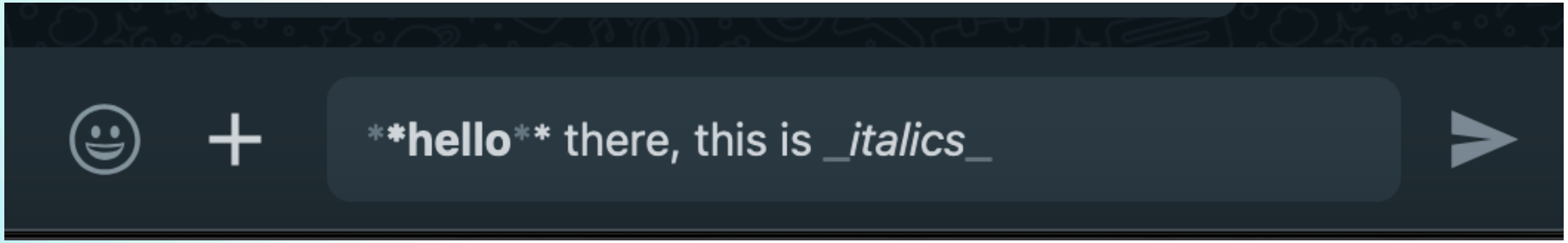
what is markdown? explain like i'm 5 year old

Markdown is like a simple way of telling your computer how to make words look nice. Imagine writing a letter, but you want some words to be **super bold**, some *a little slanty*, or even create a list of your favorite things.

Instead of drawing or using buttons, you just type special symbols while writing. For example:

- If you want something bold, you put **stars** around it: `**bold**`
- For italics, you use one star: `*italics*`
- To make a list, you just use a dash: `- Item`

When you're done, the computer looks at your text and turns it into a clean, fancy document! It's like magic writing for grown-ups who want their text to look nice. 😊


hello there, this is _italics_

https://openai.com/index/chatgpt/

Fix code | Home security | Fermat's Little Theorem | Neighbor introduction

User

what is the fermat's little theorem

ChatGPT

Fermat's Little Theorem states that if p is a prime number and a is an integer not divisible by p, then $a^p a$ (a to the power p) is congruent to a modulo p. In other words, if we divide a^p by p, the remainder is always a.For example, if $p = 5$ and $a = 2$, then $2^5 = 32$, and 32 divided by 5 has a remainder of 2. Therefore, 2^5 is congruent to 2 modulo 5, and Fermat's Little Theorem holds for this case.
Fermat's Little Theorem is often used in cryptography and other applications where it is necessary to perform modular arithmetic operations quickly and efficiently. It is also a useful tool for proving other theorems in number theory

User

how is it used in cryptography?

ChatGPT

Fermat's Little Theorem is used in cryptography in several ways. One of the most common applications is in the generation of so-called "public-key" cryptography systems, which are used to securely transmit messages over the internet and other networks. In a public-key cryptography system, each user has a pair of keys: a public key, which is widely known and can be used by anyone to encrypt a message intended for that user, and a private key, which is known only to the user and is used to decrypt messages that have been encrypted with the corresponding public key.
One way to generate these keys is to use prime numbers and Fermat's Little Theorem. For example, suppose we want to generate a public-key cryptography system for a user with the initials "ABC". We might choose two large prime numbers, p and q, and then compute the product $n = pq$. The user's public key would then be the pair (n, a), where aa is any integer not divisible by p or q. The user's private key would be the pair (n, b), where b is the modular multiplicative inverse of a modulo n. This means that when we multiply a and b together, the result is congruent to 1 modulo n. To encrypt a message with the user's public key (n, a), we first convert the message into a number m (using some agreed-upon scheme), and then compute the encrypted message c as $c = m^a \bmod n$. To decrypt the message, the recipient simply computes $m = c^b$

create systems performing tasks that typically require human intelligence.

- AI powers technologies like natural language processing, computer vision, and decision-making systems.
- Advanced AI models, such as ChatGPT and GPT-4, have billions of parameters, enabling them to generate human-like responses.
- Machine learning (a subset of AI) uses algorithms to identify patterns in data and make predictions.
- Deep learning, inspired by neural networks in the brain, has revolutionized areas like image recognition and autonomous vehicles.
- AI has significant applications across industries:
  - **Healthcare**: Assists in diagnosing diseases.
  - **Finance**: Detects fraudulent activities.
- Ethical challenges in AI include:
  - Bias in AI models.
  - Concerns about job displacement.
- Responsible AI development is crucial to addressing these challenges.

can you make this in structured markdown format

Artificial Intelligence Overview

What is AI?

Artificial Intelligence (AI) is a rapidly advancing field combining computer science and data to create systems capable of performing tasks that typically require human intelligence.

Key Capabilities

- Powers technologies like:
  - Natural language processing (e.g., ChatGPT, GPT-4).
  - Computer vision.
  - Decision-making systems.
- Advanced AI models (e.g., GPT-4) have billions of parameters, enabling them to generate human-like responses.

Subfields of AI

Machine Learning

- Uses algorithms to:
  - Identify patterns in data.
  - Make predictions.

Deep Learning

- Inspired by neural networks in the human brain.

how can i create slides from markdown?

You can easily create slides from Markdown using several tools. Below are a few popular methods:

---

1. Marp (Markdown Presentation Ecosystem)

Marp is a popular tool for converting Markdown into slides. It has a simple and efficient way to structure and style your presentations.

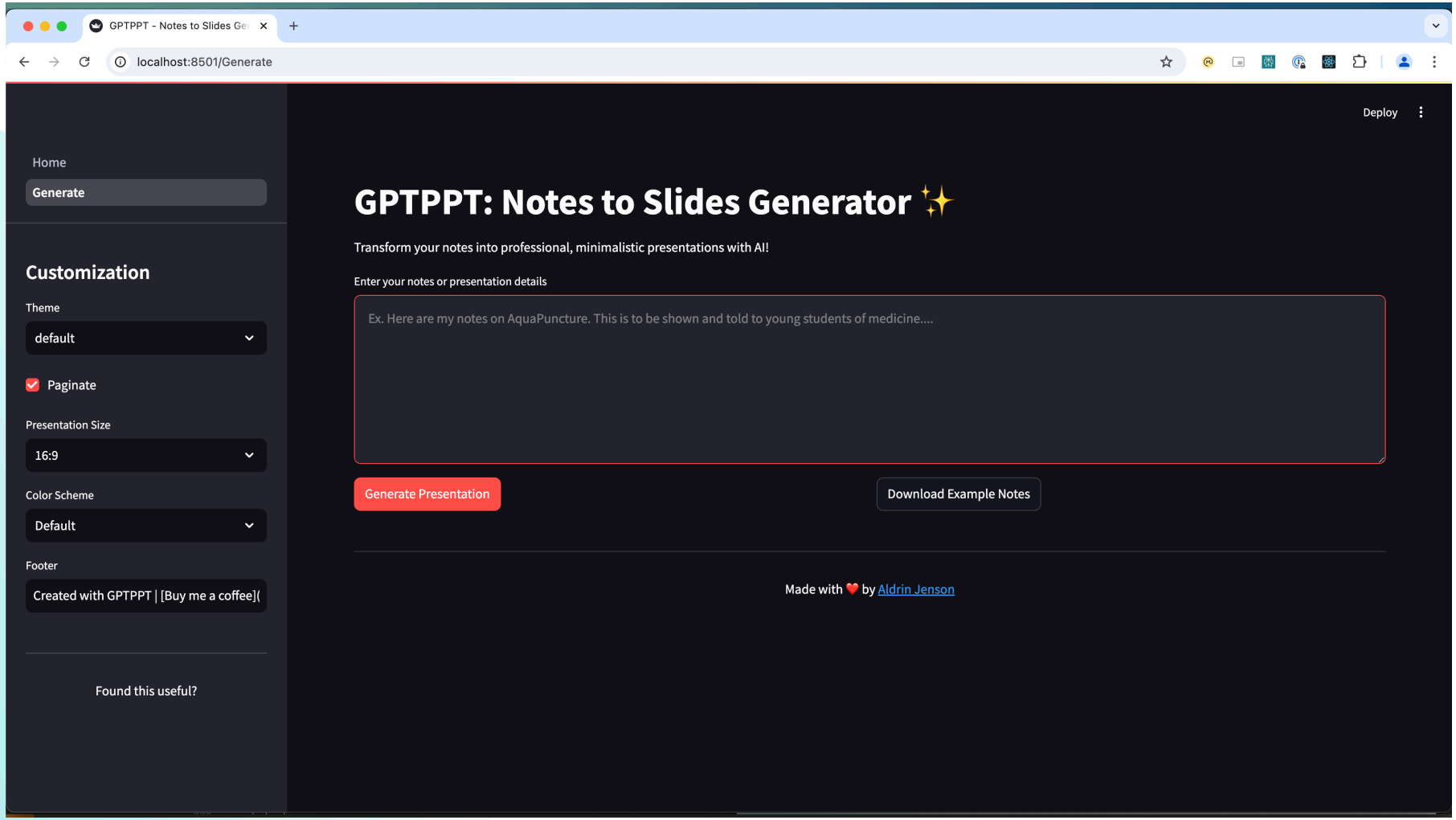
Steps:

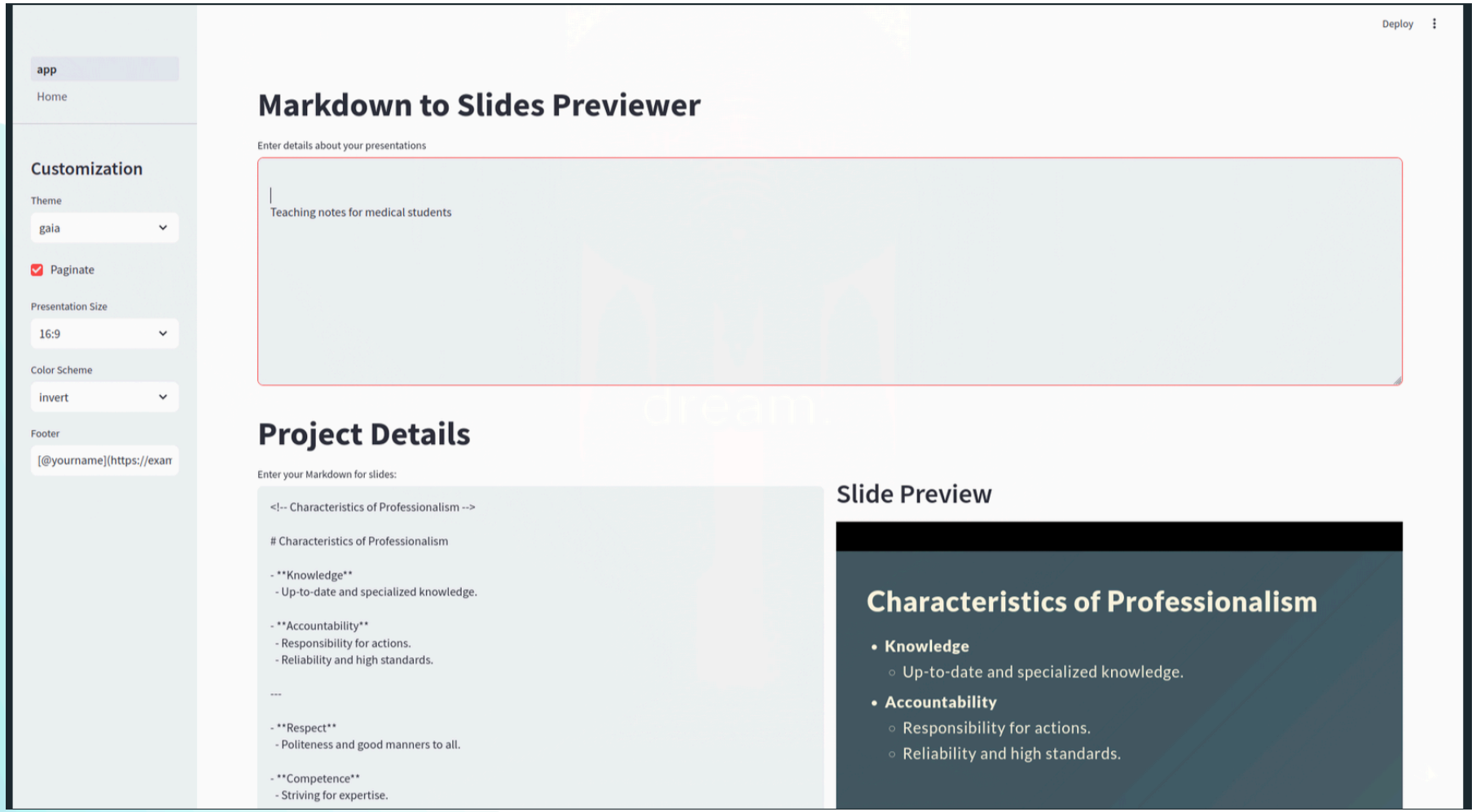
1. **Install Marp**:
   - **For VS Code**: Install the Marp for VS Code extension.
   - **Online editor**: You can also use the Marp Web Editor.
2. **Write Markdown with Slide Separators**: In Marp, use `---` to separate slides. Here's an example based on your content:

markdown Copy code

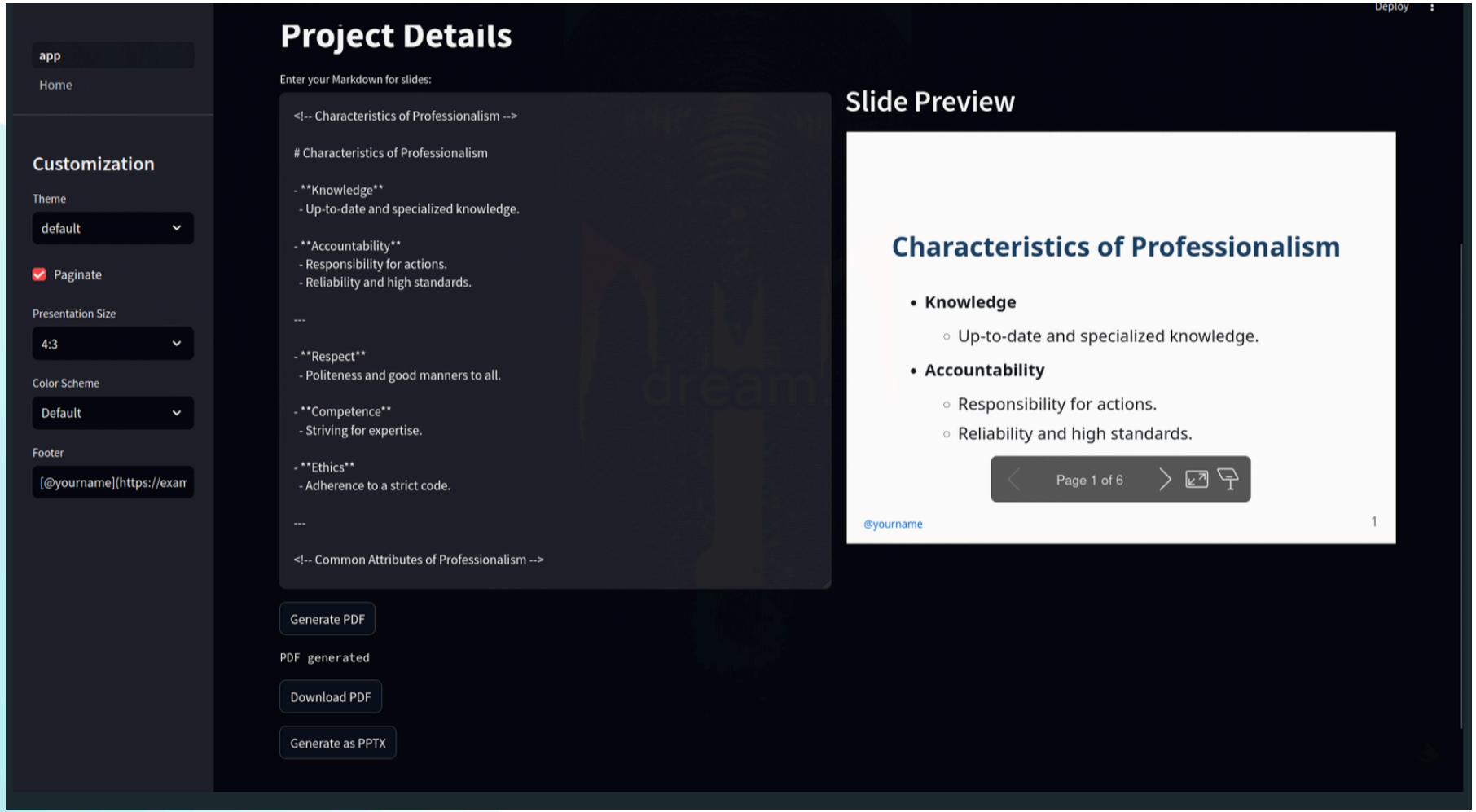
```
---
title: Artificial Intelligence Overview
---

# What is AI?
Artificial Intelligence (AI) is ↓ apidly advancing field combining computer sc
```


GPTPPT - Notes to Slides Ge
localhost:8501/Generate
Deploy
Home
Generate
Customization
Theme
default
Paginate
Presentation Size
16:9
Color Scheme
Default
Footer
Created with GPTPPT | [Buy me a coffee](
Found this useful?
GPTPPT: Notes to Slides Generator ✨
Transform your notes into professional, minimalistic presentations with AI!
Enter your notes or presentation details
Ex. Here are my notes on AquaPuncture. This is to be shown and told to young students of medicine....
Generate Presentation
Download Example Notes
Made with ❤️ by Aldrin Jenson


Deploy
app
Home
Customization
Theme
gaia
Paginate
Presentation Size
16:9
Color Scheme
invert
Footer
[@yourname](https://exam
Markdown to Slides Previewer
Enter details about your presentations
Teaching notes for medical students
Project Details
Enter your Markdown for slides:
<!-- Characteristics of Professionalism -->
Characteristics of Professionalism
- **Knowledge**
- Up-to-date and specialized knowledge.
- **Accountability**
- Responsibility for actions.
- Reliability and high standards.

- **Respect**
- Politeness and good manners to all.
- **Competence**
- Striving for expertise.
Slide Preview
Characteristics of Professionalism
• Knowledge
◦ Up-to-date and specialized knowledge.
• Accountability
◦ Responsibility for actions.
◦ Reliability and high standards.


app
Home
Customization
Theme
default
Paginate
Presentation Size
4:3
Color Scheme
Default
Footer
[@yourname](https://exam
Project Details
Enter your Markdown for slides:
<!-- Characteristics of Professionalism -->
Characteristics of Professionalism
- **Knowledge**
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- Responsibility for actions.
- Reliability and high standards.

- **Respect**
- Politeness and good manners to all.
- **Competence**
- Striving for expertise.
- **Ethics**
- Adherence to a strict code.

<!-- Common Attributes of Professionalism -->
Generate PDF
PDF generated
Download PDF
Generate as PPTX
Slide Preview
Characteristics of Professionalism
Knowledge
Up-to-date and specialized knowledge.
Accountability
Responsibility for actions.
Reliability and high standards.
Page 1 of 6
@yourname
1

Recap

User

write a limerick about it

ChatGPT

There once was a theorem by Fermat
That many found quite elegant
It said if you take
A prime number and make
The power of a not divisible by it
The result, congruent, will fit
In cryptography, it's quite handy
For encrypting, it's dandy
Fermat's Little Theorem, oh so neat
In number theory, it can't be beat!

User

summarize the conversation so far

ChatGPT

Fermat's Little Theorem states that if pp is a prime number and aa is an integer not divisible by p, then a^p (aa to the power pp) is congruent to aa modulo p. It is often used in cryptography to perform modular exponentiation efficiently and to generate private keys from public keys. It is also a useful tool in number theory for proving other theorems. I also wrote a limerick about Fermat's Little Theorem.

How can AI help?

Chatgpt can *understand* language!

Steps

- input notes to chatgpt
- use chatgpt to extract key points in markdown
- pass in markdown to traditional marp
- build some nice ui around it to make it look nice

References

- chatgpt
- marp
- streamlit
- python

Realization: Enhances / complements your skills!

Breathe...

Bbbut this involves programming!

Problem statement 2

Claude

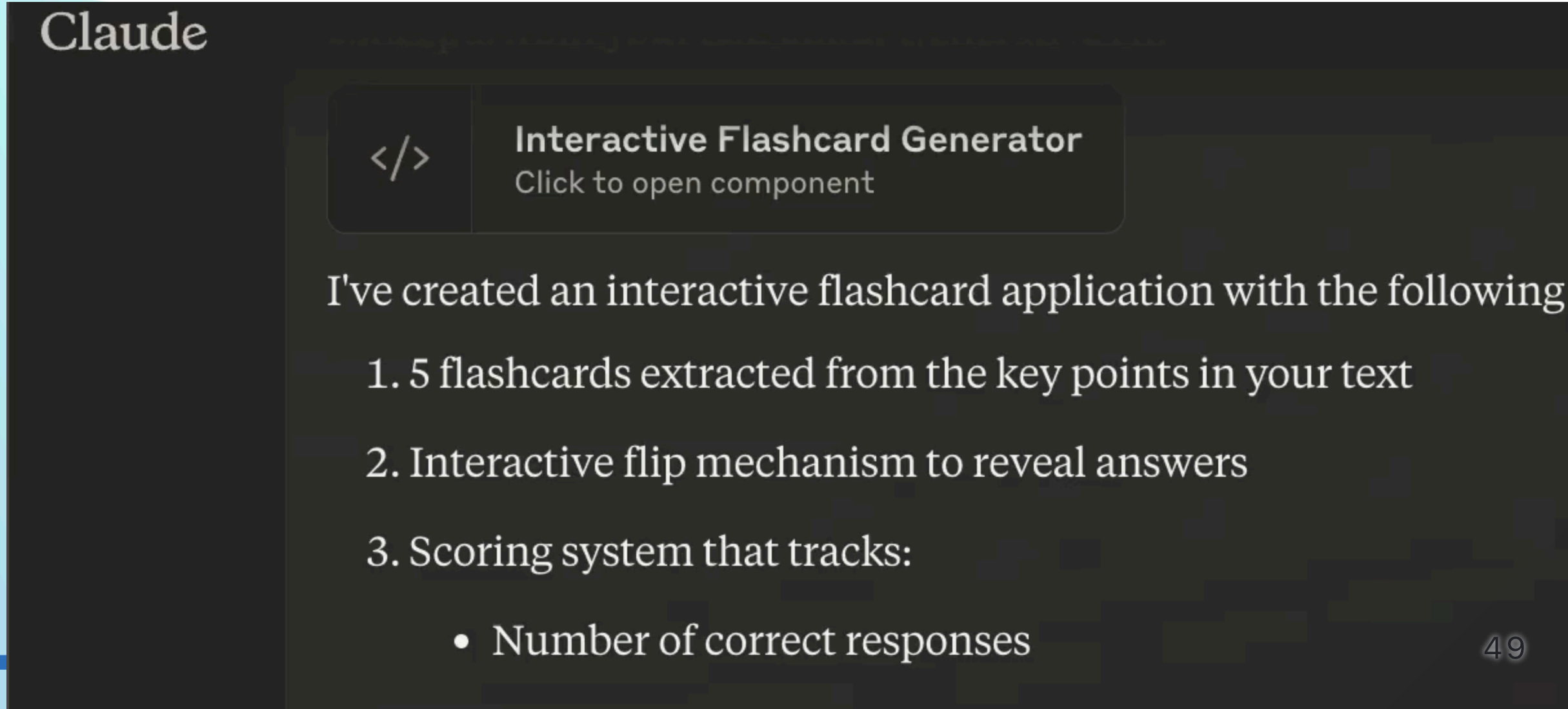
AJ

create a flash card generator. i will insert a paragraph of text from which i need you to extract key points and pass them down to the flashcard generator with questions on one side and answers on the other. keep track of scores too.

Here is the paragraph:
Generative AI refers to a branch of artificial intelligence that focuses on creating new content, such as text, images, music, and videos, based on existing data. Unlike traditional AI, which primarily classifies or predicts, generative AI learns patterns in data and uses them to produce novel outputs. Technologies like deep learning, specifically neural networks such as Generative Adversarial Networks (GANs) and Transformer models, power this capability. Popular applications include AI-driven art, chatbots, text-to-image generation, and even drug discovery. However, it raises ethical concerns such as plagiarism, misinformation, and biases embedded in training data, making responsible development crucial.

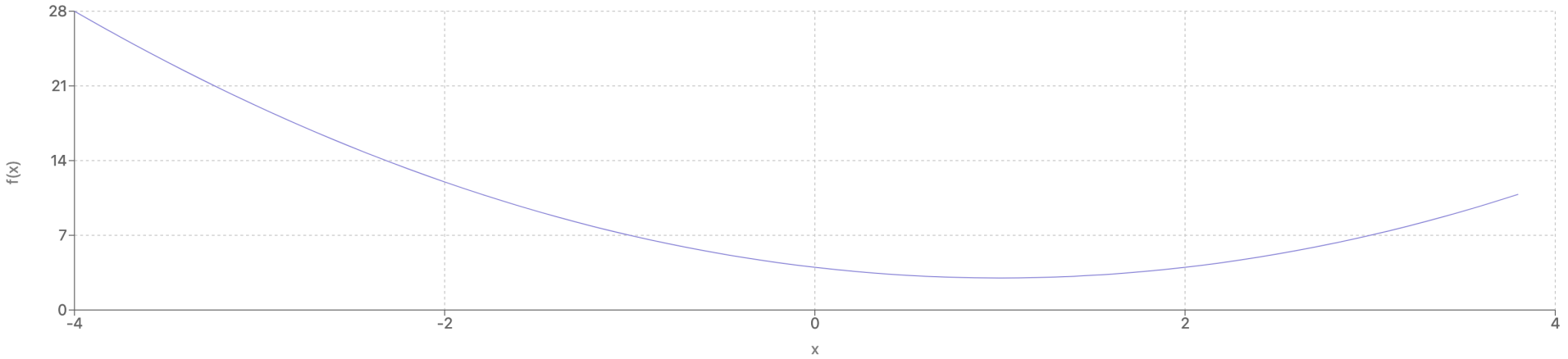
I'll help create a flash card application with a scoring system to help with learning the key concepts from your text about Generative AI.

Flashcards


Claude
Interactive Flashcard Generator
Click to open component
I've created an interactive flashcard application with the following
1. 5 flashcards extracted from the key points in your text
2. Interactive flip mechanism to reveal answers
3. Scoring system that tracks:
• Number of correct responses

Gradient Descent Visualization

Function: f(x) = x² - 2x + 4



Explanation:

- Red dots show the steps taken by gradient descent
- Starting point: x = 3.5
- Learning rate: 0.1
- The algorithm converges towards the minimum at x = 1

https://claude.ai/chat/d586f82c-baec-4842-91f2-014985a15371

Realization: Independence!

Suno

https://suno.com/

കണ്ണീർപ്പൂവിന്റെ കവിളില് തലോടി
ഈണം മുഴങ്ങും പഴമ്പാട്ടില് മുങ്ങി
കണ്ണീർപ്പൂവിന്റെ കവിളില് തലോടി
ഈണം മുഴങ്ങും പഴമ്പാട്ടില് മുങ്ങി

മറുവാക്കു കേള്ക്കാന് കാത്തു നില്ക്കാതെ
പൂത്തുമ്പിയെന്തേ മറഞ്ഞൂ എന്തേ
പുള്ളോർക്കുടം പോലെ തേങ്ങി

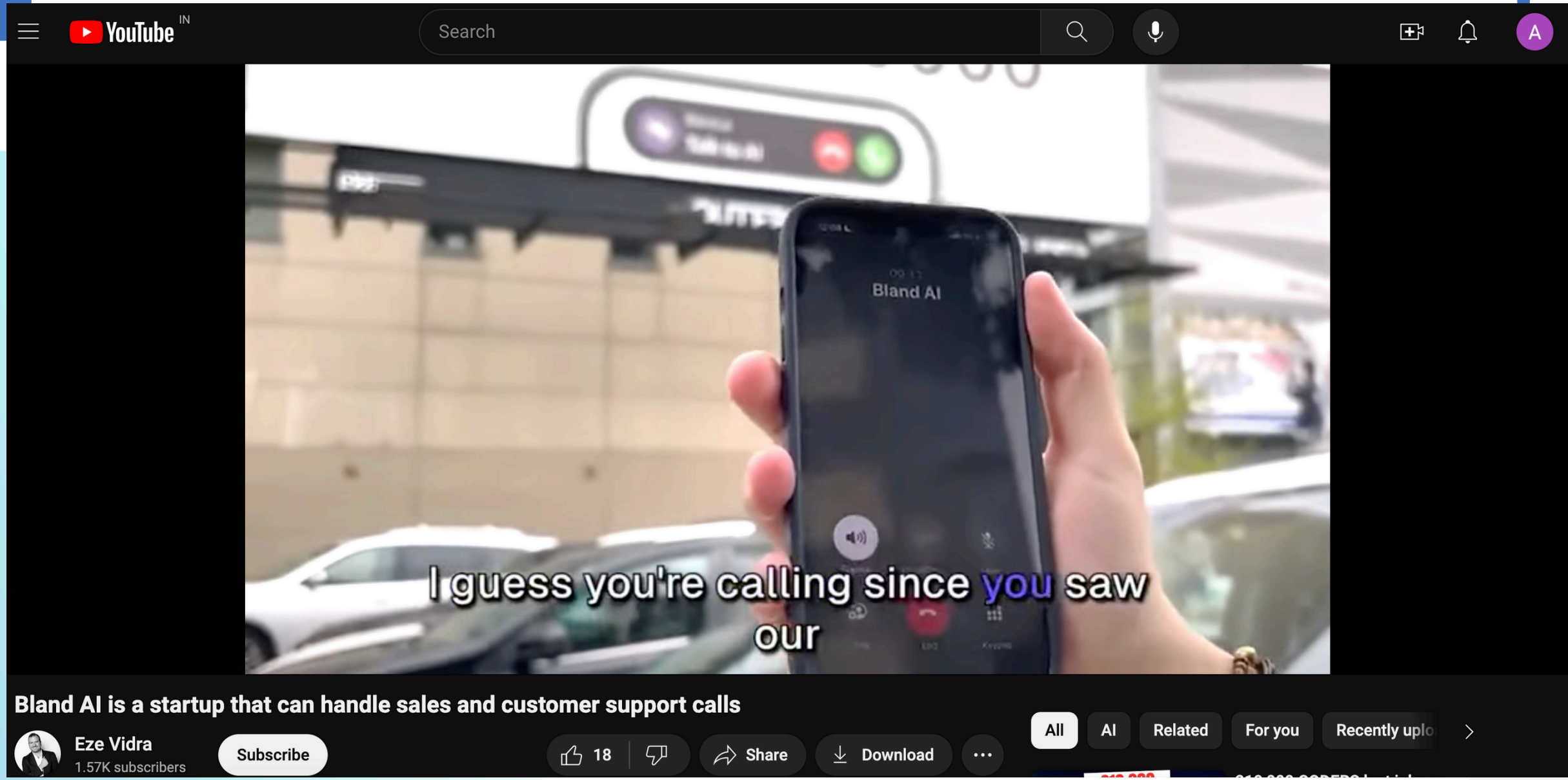
കണ്ണീർപ്പൂവിന്റെ കവിളില് തലോടി
ഈണം മുഴങ്ങും പഴമ്പാട്ടില് മുങ്ങി
ഉണ്ണിക്കിടാവിന്നു നല്കാന്
അമ്മ നെഞ്ചില് പാലാഴിയേന്തി

ആയിരം കൈ നീട്ടി നിന്നു
സൂര്യതാപമായ് താതന്റെ ശോകം
വിട ചൊല്ലവേ നിമിഷങ്ങളില്
ജലരേഖകള് വീണലിഞ്ഞൂ
കനിവേകുമീ വെണ്മേഘവും
മഴനീർക്കിനാവായ് മറഞ്ഞു ദൂരെ
പുള്ളോർക്കുടം കേണുറങ്ങി

0:00 / 2:00

0:00 / 2:00

Few other crazy things people have been doing with AI


YouTube IN
Search
Bland AI
I guess you're calling since you saw our
Bland AI is a startup that can handle sales and customer support calls
Eze Vidra
1.57K subscribers
Subscribe
18
Share
Download
All
AI
Related
For you
Recently uplo

5

conclusion:

- lessons learned
  slide gnerators + ai flash cards
  - ai complements your skills and make you more efficient
- using claude artifacts:
  - makes you independent
- chatpdf and ocr
  - make you productive with research

conclusion

- ai makes you efficient
- ai makes you independent
- ai gives you entertainment
- ai provides companionship
- ai is scary

what does this remind you of?

ANS: the internet...

living in best times.

we need to take leverage of it better.

no other best time to leverage ideas.

key idea: ai makes it possible now for us to realized our ideas.

this is the best time to build!

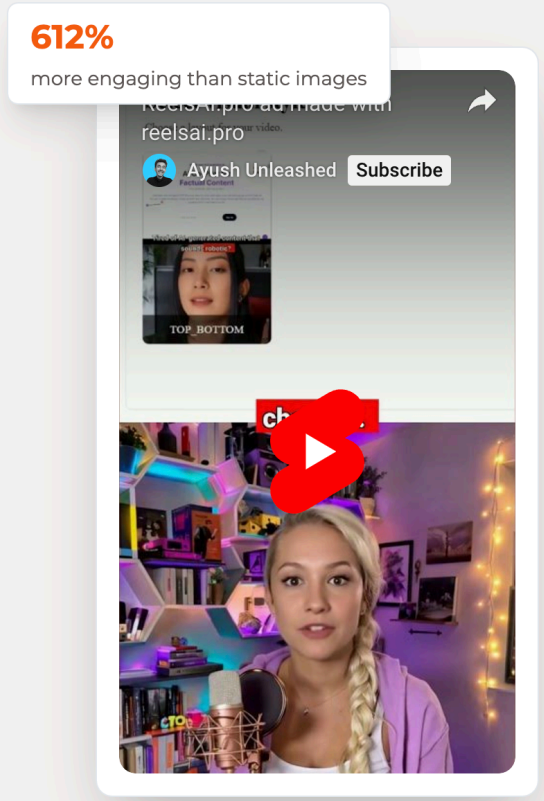
🚀 TRUSTED BY STARTUPS Currently in Beta

Create Video Ads Using AI

Transform your product into compelling video ads with the power of AI. Fast, efficient, and professionally crafted.

Get Started →

We are Currently in Beta. Get Early Access Now!



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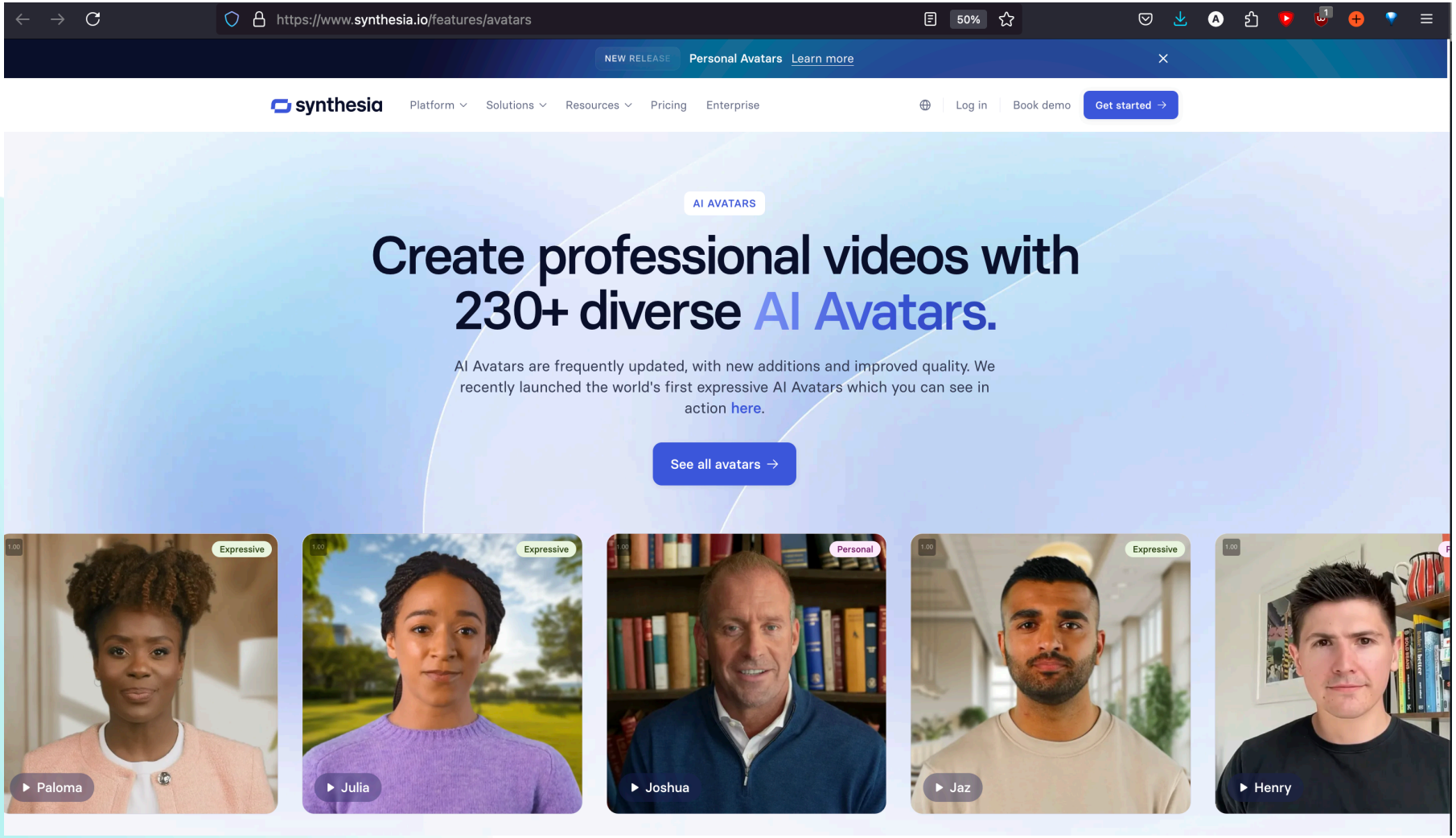

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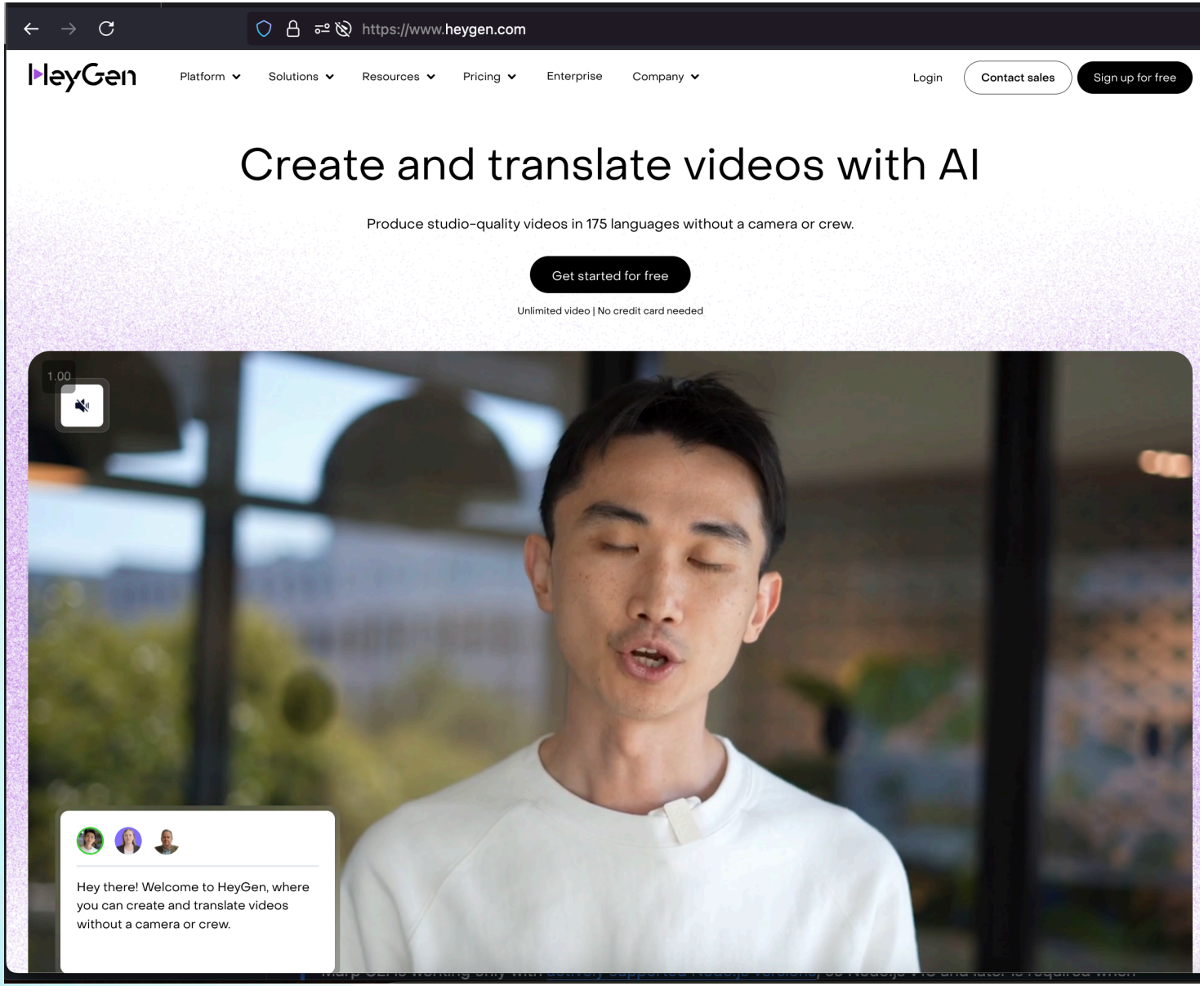

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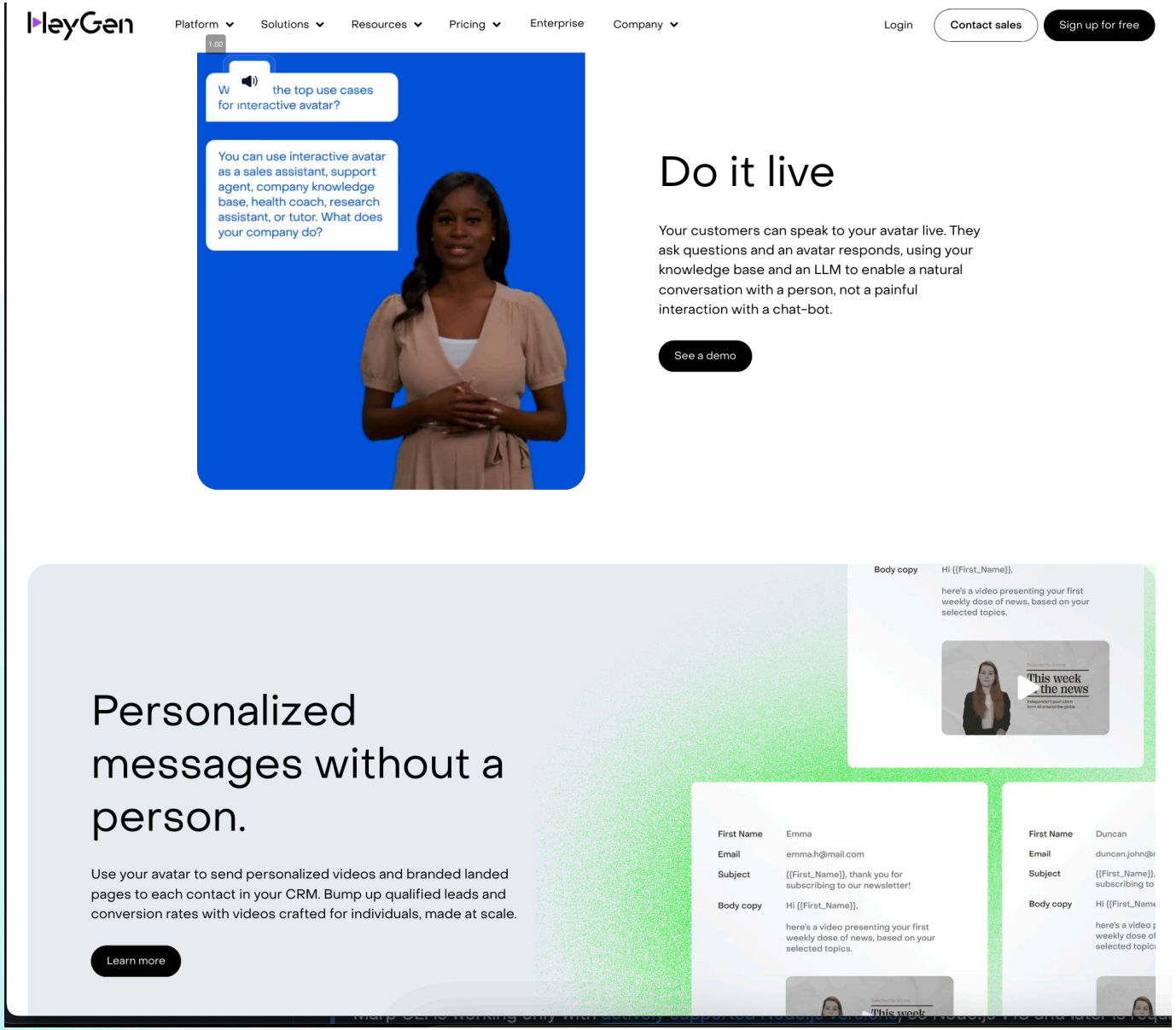
HeyGen
Platform
Solutions
Resources
Pricing
Enterprise
Company
Login
Contact sales
Sign up for free
the top use cases for interactive avatar?
You can use interactive avatar as a sales assistant, support agent, company knowledge base, health coach, research assistant, or tutor. What does your company do?
Do it live
Your customers can speak to your avatar live. They ask questions and an avatar responds, using your knowledge base and an LLM to enable a natural conversation with a person, not a painful interaction with a chat-bot.
See a demo
Personalized messages without a person.
Use your avatar to send personalized videos and branded landed pages to each contact in your CRM. Bump up qualified leads and conversion rates with videos crafted for individuals, made at scale.
Learn more
Body copy
Hi {{First_Name}},
here's a video presenting your first weekly dose of news, based on your selected topics.
First Name
Emma
Email
emma.h@mail.com
Subject
{{First_Name}}, thank you for subscribing to our newsletter!
Duncan

kristerka @kristerka · Oct 25, 2021
Great work by the @CadburyWorld in India! 'Not just a Cadbury ad': Shah Rukh Khan turns brand ambassador for local businesses. They used machine learning to recreate the actor's face and voice to promote local stores. #MachineLearning #Advertising #ArtificialIntelligence
onics
Podja Electronics
Shivam
10515
400710
Different versions of the same ad with local stores
0:03 / 0:45

Thank You ✨

Resources